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Healthcare AI

Intended and Unintended Uses

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Document Control

This document is the responsibility of the Author. It is subject to formal change control after the initial approved release (e.g., version 1.0). Change requests must be communicated to the Author in writing. Agreed changes are recorded in the Version History of the document.

Version History

Date	Version	Author	Changes
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Reviewers

Name	Role	Date
Jonathan Doe	Head of AI Development	01/12/20XX

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Background

Healthcare AI provides patient diagnoses and medical research support to enable better patient outcomes and research data analysis. The system employs deep learning techniques (specifically convolutional neural networks) for natural language processing, extraction of information from text data and interpretation of available data leading to a diagnosis prediction.

Intended and Unintended Uses

Intended Uses

- **Medical device classification**: Healthcare AI is classified as Class III per the EU Medical Device Risk classification.
- **Medical research and patient diagnoses support**: Healthcare AI is intended for use by healthcare professionals within customer organizations, to support research into medical conditions and patient diagnoses. Output is advisory and for interpretation by healthcare professionals, who relate information (such as diagnoses) to patients.
- **Diagnoses support**: Healthcare AI can diagnose a broad spectrum of medical conditions based on third party sourced medical information, as well as from the medical research that it also supports, thus learning from the healthcare professionals as they undertake research. Healthcare AI can also learn from any modification to its diagnoses by healthcare professionals through continuous learning.
- **AI subject coverage**: Healthcare AI is intended for use for any patient and by any healthcare professional, regardless of ethnic or cultural background, and supports multiple natural languages aligned to the geographical regions in which it is in use.
- **High availability**: The system is available to healthcare professionals 24/7 through various edge devices to support patient diagnoses in customer organization locations (such as workstations in GP surgeries) or the field (using mobile devices or the on-line portal from patient locations).
- **Access to medical information**: Healthcare AI provides a secure front-end to the models within the <<organization>> Predictive Analytics Engine, employing authentication techniques as provided in the edge devices (multi-factor authentication, biometrics, etc.).
- **Management of medical research**: Medical research is intended for use and publication by healthcare professionals, requiring interpretation prior to publication.

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- **Continuous learning**: Medical research, patient details and diagnoses information is fed back through continuous learning for use in training, validation and test data.
- **Regulatory support**: Healthcare AI provides transparency in support of medical regulatory requirements and investigations.
- **Demographics support**: Healthcare AI can provide anonymized summary information about medical conditions, such as numbers of patients suffering from a given condition to medical authorities, government agencies or other interested parties.

Unintended Uses

- **Use by patients**: The system is not intended for use by the general public/patients directly. Patients' unauthorized use of the system could provide 'unfiltered' patient diagnoses directly to patients. Without the interpretation of a healthcare professional, diagnoses could be misunderstood or cause distress to patients.
- **Human oversight**: Healthcare professionals are expected to provide oversight of diagnoses to ensure accuracy and patient confidence.
- **Unauthorized access to medical information**: Unauthorized access to Healthcare AI, or the back-end Predictive Analysis Engine would allow access to sensitive information about patients and their medical conditions/diagnoses, as well as sensitive information about healthcare professionals.
- **Unauthorized release of medical information**: Release of medical information without patient approval, for example to insurance providers, government agencies or to patient employers could have a detrimental impact on the patient's career, wider lifestyle and mental state.
- **Undermining of established peer review processes**: Publishing of medical research directly from Healthcare AI outside of established peer review processes within the medical profession could undermine said processes and patient confidence.
- **Undermining patient confidentiality**: Exploitation of transparency could lead to medical information related to individuals being obtained or inferred.
- **Automated issuing of prescriptions**: Healthcare AI is not intended to issue prescriptions for dispensing to patients without receiving the oversight by healthcare professionals to provide assurance to patients. However, this could be seen as a logical next step.

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- **Automated patient diagnoses**: A desirable outcome of continuous learning is to refine the model sufficiently so that diagnoses can be delivered directly to patients without needing interpretation by a healthcare professional. This is anticipated to require regulatory approval and some form of multi-jurisdictional certification to be acceptable. As such, this is seen as a long-term goal but is not an intended use at this time.

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